

ALTERNATE GENERATION.

In several insect families, the species is not wholly represented in the adult individuals of both sexes, or in their development, but, to complete this series, supplementary individuals, as it were, of one or of several preceding generations, are required. The son may not resemble the father, but the grandfather, and in some instances, the likeness re-appears only in latter generations. Agassiz states: "Alternate generation was first observed among the Salpæ. These are marine mollusks, without shells, belonging to the family Tunicata. They are distinguished by the curious peculiarity of being united together in considerable numbers so as to form long chains, which float in the sea, the mouth (*m*) however being free in each.

Fig. 2.

Fig. 3.



"Fig. 2. The individuals thus joined in floating colonies produce eggs; but in each animal there is generally but one egg formed, which is developed in the body of the parent, and from which is hatched a little mollusk.

"Fig. 3, which remains solitary, and differs in many respects from the parent. This little animal, on the other hand, does not produce eggs, but propagates, by a kind of budding, which gives rise to chains already seen in the body of their parent (*a*), and these again bring forth solitary individuals, etc."

It therefore follows that generation in some animals requires two different bodies with intermediate ones, by means of which, and their different modes of reproduction, a return to the original stock is effected.

Universality of Animalcular Life.—Living organisms are universally diffused over every part of the globe. The gentle zephyr wafts from flower to flower invisible, fructifying atoms, which quicken beauty and fragrance, giving the promise of a golden fruitage, to gladden and nourish a dependent world. Nature's own sweet cunning invests all living things,

constraining into her service chemical affinities, arranging the elements and disposing them for her own benefit, in such numberless ways that we involuntarily exclaim,

“The course of Nature is the art of God.”

The microscope reveals the fact that matter measuring only $\frac{1}{125000}$ of an inch in diameter may be endowed with vitality, and that countless numbers of animalcules often inhabit a single drop of stagnant water. These monads do not vary in form, whether in motion or at rest. The life of one, even, is an inexplicable mystery to the philosopher. Ehrenberg writes: “Not only in the polar regions is there an uninterrupted development of active microscopic life, where larger animals cannot exist, but we find that those minute beings collected in the Antarctic expedition of Captain James Ross exhibit a remarkable abundance of unknown, and often most beautiful forms.”

Even the interior of animal bodies is inhabited by animalcules. They have been found in the blood of the frog and the salmon, and in the optic fluid of fishes. Organic beings are found in the interior of the earth, into which the industry of the miner has made extensive excavations, sunk deep shafts, and thus revealed their forms; likewise, the smallest fossil organisms form subterranean strata many fathoms deep. Not only do lakes and inland seas abound with life, but also, from unknown depths, in volcanic districts, arise thermal springs which contain living insects. Were we endowed with a microscopic eye, we might see myriads of ethereal voyagers wafted by on every breeze, as we now behold drifting clouds of aqueous vapor. While the continents of earth furnish evidences of the universality of organic beings, recent observations prove that “animal life predominates amid the eternal night of the depths of the liquid ocean.”

THE ORIGIN OF LIFE.

The ancients, rude in many of their ideas, referred the origin of life to divine determination. The thought was crudely expressed, but well represented, in the following verse:

“Then God smites his hands together,
And strikes out a soul as a spark,
Into the organized glory of things,
From the deeps of the dark.”

According to a Greek myth, Prometheus formed a human image from the dust of the ground, and then, by fire stolen from heaven, animated it with a living soul. Spontaneous generation once held its sway, and now the idea of natural evolution is popular. Some believe that the impenetrable mystery of life is evolved from the endowments of nature, and build their imperfect theory on observations of her concrete forms and their manifestations, to which all our investigations are restricted. But every function indicates purpose, every organism evinces intelligent design, and *all* proclaim a Divine Power. Something cannot come out of nothing. With reason and philosophy, *chance* is an impossibility. We, therefore, accept the display of wisdom in nature as indicative of the designs of God. Thus "has He written His claims for our profoundest admiration and homage all over every object that He has made." If you ask: Is there any advantage in considering the phenomena of nature as the result of DIVINE VOLITION? we answer, that this belief corresponds with the universally acknowledged ideas of accountability; for, with a wise and efficient Cause, we infer there is an intelligent creation, and the desire to communicate, guide and bless, is responded to by man, who loves, obeys, and enjoys. Nothing is gained by attributing to nature vicegerent forces. Is it not preferable to say that she responds to intelligent, loving Omnipotence? Our finiteness is illustrated by our initiation into organized being. Emerging from a rayless atom, too diminutive for the sight, we gradually develop and advance to the maturity of those *conscious powers*, the exercise of which furnishes indubitable evidence of our immortality. We are pervaded with invisible influences, which, like the needle of the compass trembling on its pivot, point us to immortality as our ultimate goal, where in the sunny clime of Love, even in a spiritual realm of joy and happiness, we may eternally reign with Him who is all in all.

CHAPTER II.

PHYSIOLOGICAL ANATOMY.

THE BONES.

All living bodies are made up of tissues. There is no part, no organ, however soft and yielding, or hard and resisting, which has not this peculiarity of structure. The *bones* of animals, as well as their flesh and fat, are composed of tissues, and all alike made up of cells. When viewed under a microscope, each cell is seen to consist of three distinct parts, a *nucleolus*, or dark spot, in the center of the cell, around which lies a mass of granules, called the *nucleus*; and this, in turn, is surrounded with a delicate, transparent membrane, termed the *envelope*. Each of the granules composing the nucleus assimilates nourishment, thereby growing into an independent cell, which possesses a triple organization similar to that of its parent, and in like manner reproduces other cells.

A variety of tissues enters into the composition of an animal structure, yet their differences are not always distinctly marked, since the characteristics of some are not unlike those of others. We shall notice, however, only the more important of the tissues.

The *Areolar*, or *Connective Tissue*, is a complete network of delicate fibers, spread over the body, and serves to bind the various organs and parts together. The fibrous and serous tissues are modifications of the areolar.

Fig. 4.



Nucleated cell.
From Goeber. 1. Periphery of the cell, or cell-wall. 2. Nucleus. 3. Nucleolus in the center.

The *Nervous Tissue* is of two kinds: The gray, which is pulpy and granulated, and the white fibrous tissue. The *Adipose Tissue* is an extremely thin membrane, composed of closed cells which contain fat. It is found principally just beneath the skin, giving it a smooth, plump appearance.

Fig. 5.



Arrangement of fibers in the Areolar Tissue. Magnified 135 diameters.

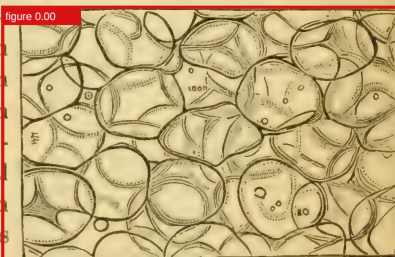
The *Cartilaginous Tissue* consists of nucleated cells, and, with the exception of bone, is the hardest part of the animal frame. The *Osseous Tissue*, or bone, is more compact and solid than the cartilaginous, for it contains a greater quantity of lime. The *Muscular*

Tissue is composed of bundles of fibers, which are enclosed in a cellular membrane.

Various opinions have been entertained in regard to the formation, or growth, of bone. Some anatomists have supposed that all bone is formed in cartilage. But this is not

true, for there is an *intra-membranous*, as well as an *intra-cartilaginous*, formation of bone, as may be seen in the development of the cranial bones, where the gradual calcification takes place upon the inner layers of the fibrous coverings. Intra cartilaginous deposit is found in the vicinity

Fig. 6.

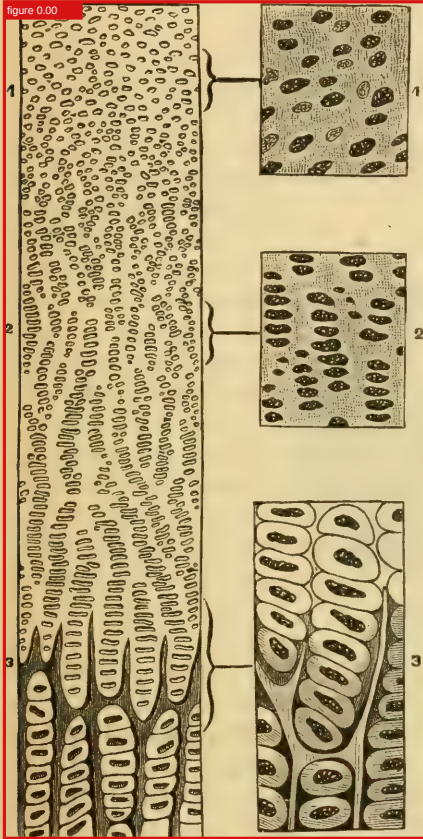


Human Adipose Tissue.

of the blood-vessels, within the cartilaginous canals; also, there are certain points first observed in the shafts of long bones, called *centers of ossification*. These points are no sooner formed than the cartilage corpuscles arrange themselves in concentric zones, and, lying in contact with one another, become very compact. As ossification proceeds, the cup-shaped cavities are converted into closed interstices of bone, with extremely thin lamellæ, or layers. These, however, soon increase

in density, and no blood-vessels can be observed within them.

Fig. 7.



Vertical section of cartilage near the surface of ossification. 1. Ordinary appearance of the temporary cartilage. 1'. Portion of the same more highly magnified. 2. The cells beginning to form into concentric zones. 2'. Portion more magnified. 3. The ossification is extending in the intercellular spaces, and the rows of cells are seen resting in the cavities so formed, the nuclei being more separated than above. 3'. Portion of the same more highly magnified.

their ends and surfaces. In the child, their extremities are

The bony plates form

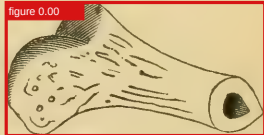
the boundaries of the Haversian, or nutritive canals of the bones. In the second stage of ossification, the cartilage corpuscles are converted into bone. Becoming flattened against the osseous lamellæ already formed, they crowd upon one another so as to entirely obliterate the lines that distinguish them; and, simultaneously with these changes, a calcareous de-

Fig. 8.
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Thigh-bone,
sawn open
lengthwise.

Fig. 9.



Lower end of the thigh-bone
sawn across, showing its central
cavity.

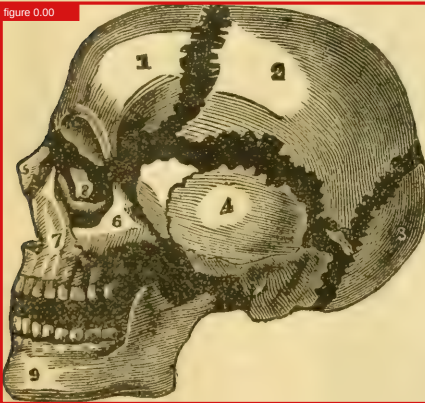
posit takes place upon their interior. Bones grow by additions to

separated from the body of the bone by a layer of cartilage, and the cancellated, or cellular structure, which remains for a time in the interior, represents the early condition of the ossifying substances.

The bones contain more earthy matter in their composition than any other part of the human body, being firm, hard, and of a lime color. They compose the skeleton or frame work, and, when united by natural ligaments, form what is known as the *natural* skeleton; when they are wired together, they are called an *artificial* skeleton.

Fig. 10.

figure 0.00



The bones of the skull separated. 1. Frontal, only half is seen. 2. Parietal. 3. Occipital, only half is seen. 4. Temporal. 5. Nasal. 6. Malar. 7. Superior maxillary (upper jaw). 8. Lachrymal. 9. Inferior maxillary (lower jaw). Between 4 and 6 a part of the sphenoid or wedge-shaped bone, is seen. Another bone assisting to form the skull, but not here seen, is called the *ethmoid* (sieve-like, from being full of holes), and is situated between the sockets of the eyes, forming the roof of the nose.

ital, two *parietal*, two *temporal*, *frontal*, *sphenoid*, and *ethmoid*. Those composing the face are, the two *nasal*, two *superior maxillary*, two *lachrymal*, two *malar* two *palate*, two *inferior turbinated*, *vomer*, and *inferior maxillary*. The cranial bones are composed of two dense plates, between which there is, in most places, a cancellated or cellular tissue. The external

The number of bones in the human body is variously estimated; for those regarded as single by some anatomists are considered by others to consist of several distinct pieces. There are two hundred distinct bones in the human skeleton besides the teeth. These may be divided into those of the Head, Trunk, Upper Extremities, and Lower Extremities.

The Bones of the Head are classed as follows: eight belonging to the Cranium, and fourteen to the Face.

The bones of the Cranium are the *occipital*,

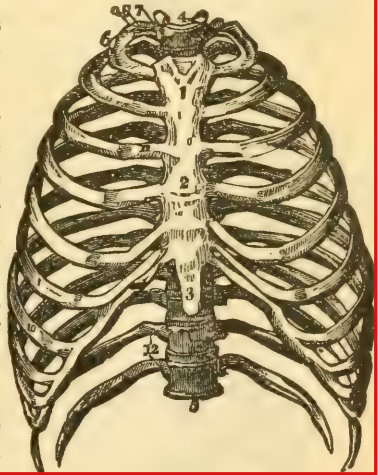
plate is fibrous, the internal, compact and vitreous. The skull is nearly oval in form, convex externally, the bone being much thicker at the base than elsewhere, and it is, in every respect admirably adapted to resist any injury to which it may be exposed, thus affording ample protection to the brain substance which it envelops. The internal surface of the cranium presents eminences and depressions for lodging the convolutions of the brain, and numerous furrows for the ramifications of the blood-vessels. The bones of the cranium are united to one another by ragged edges called *sutures*, which are quite distinct in the child, but which in old age are nearly effaced. Some authorities suppose that by this arrangement the cranium is less liable to be fractured by blows; others think that the sutures allow the growth of these bones, which takes place by a gradual osseous enlargement at the margins. The bones of the *Face* are joined at the lower part and in front of the cranium, and serve for the attachment of powerful muscles which assist in the process of mastication. Although the soft parts of the face cover the bony structure, yet they do not conceal its principal features, or materially change its proportions. The form of the head and face presents some remarkable dissimilarities in different races.

The Trunk has fifty-four

bones, which are as follows: The *Os Hyoides*, the *Sternum*, twenty-four Ribs, twenty-four *vertebræ* or bones of the Spinal Column, the *Sacrum*, the *Coccyx*, and two *Ossa Innominata*. The *Os Hyoides*, situated at the base of the tongue, is the most isolated bone of the skeleton, and serves for the attachment of

figure 0.00

Fig. 11.



1. The first bone of the sternum (breast-bone). 2. The second bone of the sternum.
3. The cartilage of the sternum. 4. The first dorsal vertebra (a bone of the spinal column). 5. The last dorsal vertebra. 6. The first rib. 7. Its head. 8. Its neck. 9. Its tubercle. 10. The seventh or last true rib. 11. The cartilage of the third rib. 12. The floating ribs.

muscles. The *Sternum*, or breast-bone, in a child is composed of six pieces, in the adult of three, which in old age are consolidated into one bone. The *Ribs* are thin, curved bones, being convex externally. There are twelve on each side, and all are attached to the spinal column. The seven upper ribs, which are united in front of the sternum, are termed *true ribs*; the next

Fig. 12.



A vertebra of the neck. 1. The body of the vertebra. 2. The spinal canal. 4. The spinous process cleft at its extremity. 5. The transverse process. 7. The inferior articular process. 3. The superior articular process.

figure 0.00

Fig. 13.



1. The cartilaginous substance which connects the bodies of the vertebrae. 2. The body of the vertebra. 3. The spinous process. 4, 4. The transverse processes. 5, 5. The articular processes. 6, 6. A portion of the bony bridge which assists in forming the spinal canal (7).

ing one is termed the *spinous*. The transverse

three, which are not attached to the sternum, but to one another are called *false ribs*; and the last two, which are joined only to the vertebrae, are designated as *floating ribs*. The first rib is the shortest, and they increase in length as far as the eighth, after which this order is reversed.

The *Spinal Column* or back-bone, when viewed from the front presents a perpendicular appearance, but a side view shows four distinct curves. The bones composing it are called *vertebrae*. The body part of a vertebra is light and spongy in texture, having seven projections called *processes*, four of which are the *articular processes*, which furnish surfaces to join the different vertebrae of the spinal column. Two are called *transverse*, and the remain-

Fig. 14.

figure 0.00



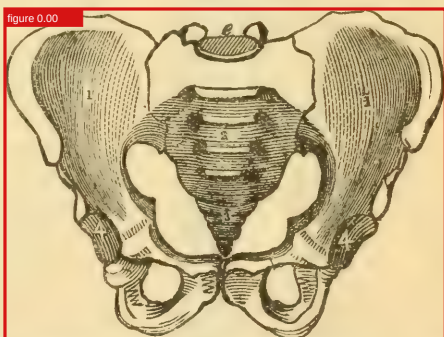
Back-bone, spinal column, or vertebral column. All animals possessing such a row of bones are called *vertebrates*. Above *b* are the cervical (neck) vertebrae; *b* to *c*, dorsal (back) or chest vertebrae; *c* to *d*, lumbar (loins) vertebrae; *d* to *e*, sacrum; *e* to *f*, coccyx.

and spinous processes serve for the attachment of the muscles belonging to the back. All these processes are more compact than the body of the vertebra, and, when naturally connected, are so arranged as to form a tube which contains the *medulla spinalis*, or spinal cord. Between the vertebræ is a highly-elastic, cartilaginous and cushion-like substance, which freely admits of motion, and allows the spine to bend as occasion requires. The natural curvatures of the spinal column diminish the shock produced by falling, running or leaping, which would otherwise be more directly transmitted to the brain. The ribs at the sides, the sternum in front, and the twelve dorsal bones of the spinal column behind, bound the thoracic cavity, which contains the lungs, heart, and large blood-vessels.

The *Pelvis* is an open bony structure, consisting of the Os Innominata, one on either side, and the Sacrum and Coccyx behind.

The *Sacrum*, during childhood, consists of five bones, which in later years unite to form one bone. It is light and spongy in texture, and the upper surface articulates with the lowest vertebra, while it is united at its inferior margin to the coccyx. The *Coccyx* is the terminal bone of the spinal column. In infancy it is cartilaginous and com-

Fig. 15.



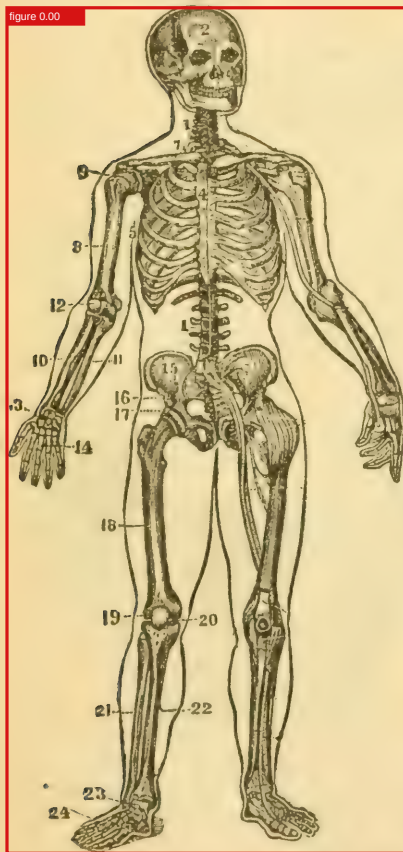
A representation of the pelvic bones. *c.* The lumbosacral joint. 2. The sacrum. 3. Coccyx. 1. The innominata. 4, 4. Acetabula.

posed of several pieces, but in the adult these unite and form one bone. The *Innominata*, or nameless bones, during youth, consist of three separate pieces on each side; but as age advances they coalesce and form one bone. A deep socket, called the *acetabulum*, is found near their junction, which serves for the reception of the head of the thigh-bone.

The Bones of the Upper Extremities are sixty-four in number, and are classified as follows: The Scapula,

Clavicle, Humerus, Ulna, Radius, Carpus, Metacarpus, and Phalanges. The *Scapula*, or shoulder-blade, is an irregular, thin,

Fig. 16.



1. Portions of the back-bone. 2. Cranial bones. 3. Breast-bone. 4. Ribs. 5. Collar-bone. 6. Arm-bone (humerus). 7. Shoulder-joint. 8, 9. Bones of the fore-arm (ulna and radius). 10. Elbow-joint. 11. Wrist-joint. 12. Bones of the hand. 13. Pelvic bones. 14. Hip-joint. 15. Femur. 16, 17. Bones of the knee-joint. 18, 19. Fibula and tibia. 20. Ankle bone. 21. Bones of the foot.

triangular bone, situated at the posterior part of the shoulder, and attached to the upper and back part of the chest. The *Clavicle*, or collar-bone, is located at the upper part of the chest, between the sternum and scapula, and connects with both. Its form resembles that of the italic letter *f*, and it prevents the arms from sliding forward. The *Humerus*, the first bone of the arm, is long, cylindrical, and situated between the scapula and fore-arm. The *Ulna* is nearly parallel with the radius, and situated on the inner side of the fore-arm. It is the longer and larger of the two bones, and in its articulation with the humerus, forms a perfect hinge-joint. The *Radius*, so called from its resemblance to a spoke, is on the outer side of the fore-arm, and articulates with the

bones of the wrist, forming a joint. The ulna and radius also articulate with each other at their extremities. The *Carpus*, or wrist, consists of eight bones, arranged in two rows. The

Metacarpus, or palm of the hand, is composed of five bones,

situated between the carpus and fingers. The *Phalanges*, fourteen in number, are the bones of the fingers and thumb, the fingers each having three and the thumb two.

The Bones of the Lower Extremities, sixty in number, are classed as follows: The Femur, Patella, Tibia, Fibula, Tarsus, Metatarsus, and Phalanges. The *Femur*, or thigh-bone, is the longest bone in the body. It has a large round head, which is received into the acetabulum, thus affording a good illustration of a ball and socket joint. The *Patella*, or knee-pan, is the most complicated articulation of the body. It is of a round form, connects with the tibia by means of a strong ligament, and serves to protect the front of the joint, and to increase the leverage of the muscles attached to it, by causing them to act at a greater angle. The *Tibia*, or shin bone, is enlarged at each extremity and articulates with the femur above and the astragalus, the upper bone of the tarsus, below. The *Fibula*, the small bone of the leg, is situated on the outer side of the tibia, and is firmly bound to it at each extremity. The *Tarsus*, or instep, is composed of seven bones, and corresponds to the carpus of the upper extremities. The *Metatarsus*, the middle of the foot, bears a close resemblance to the metacarpus, and consists of five bones situated between the tarsus and the phalanges. The tarsal and the metatarsal bones are so united as to give an arched appearance to the foot, thus imparting elasticity. The *Phalanges*, the toes, consist of fourteen bones, arranged in a manner similar to that of the fingers.

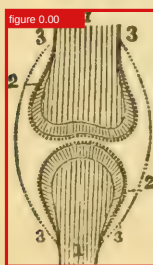
We are not less interested in tracing the formation of bone through its several stages, than in considering other parts of the human system. The formation of the Haversian canals for the passage of blood-vessels to nourish the bones, the earlier construction of bony tissue by a metamorphosis of cartilaginous substance, and also the commencement of ossification at distinct points, called *centers of ossification*, are all important subjects, requiring the student's careful attention. The bones are protected by an external membranous envelope, which, from its situation is called the *periosteum*. The bones are divided into four classes, *long*, *short*, *flat* and *irregular*, being thus adapted to subserve a variety of purposes.

The Long Bones are found in the limbs, where they act as

levers to sustain the body and aid in locomotion. Each *long* bone is composed of a cylinder, known as the *shaft*, and two *extremities*. The shaft is hollow, its walls being *thickest* in the middle and growing thinner toward the extremities. The *extremities* are usually considerably enlarged, for convenience of connection with other bones and to afford a broad surface for the attachment of muscles. The clavical, humerus, radius, ulna, femur, tibia, fibula, the bones of the metacarpus, metatarsus and the phalanges, are classed as long bones.

Where the principal object to be attained is strength, and the motion of the skeleton is limited, the individual bones are short and compressed, as the bones of the carpus and tarsus. The structure of these bones is spongy, except at the surface, where there is a thin crust of compact matter.

Fig. 17.



Anatomy of a joint. 1, 1. Bones of a joint. 2, 2. Cartilage. 3, 3, 3, 3. Synovial membrane.

Fig. 18.



Anatomy of the knee-joint.

1. Lower end of thigh-bone.
2. Knee-pan. 3, 4. Ligaments of the knee-pan. 5. Upper end of the tibia, or shin-bone.
6, 12. Cartilages.

When protection is required for the organs of the body, or a broad flat surface for the attachment of the muscles, the bones are expanded into plates, as in the cranium and shoulder-blades.

The *irregular* or *mixed* bones are those which, from their peculiar shape, cannot be classed among any of the foregoing divisions. Their structure is similar to the others, consisting of cancellar tissue, surrounded by a crust of compact matter.

The vertebræ, sacrum, coccyx, temporal, sphenoid, ethmoid, malar, two maxillary, palate, inferior turbinated, and hyoid are known as irregular bones.

The formation of the joints requires not only bones, but also

cartilages, ligaments, and the synovial membrane, to complete the articulation. *Cartilage* is a smooth, elastic substance, softer than bone, and invested with a thin membrane, called *perichondrium*. When cartilage is placed upon convex surfaces, the reverse is true. The *Ligaments* are white, inelastic, tendinous substances, softer than cartilage, but harder than membrane. Their function is to bind together the bones. The *Synovial Membrane* covers the cartilages, and is then reflected upon the ligaments, thus forming a thin, closed sac, called the *synovial capsule*.

All the synovial membranes secrete a lubricating fluid, termed *synovia*, which enables the surfaces of the bones and ligaments to move freely upon one another. When this fluid is secreted in excessive quantities, it produces a disease known as "dropsy of the joints." There are numerous smaller sacs besides the synovial, called *bursæ mucosæ*, which in structure are analogous to them, and secrete a similar fluid. Some joints permit motion in every direction, as the shoulders, some in two directions only, as the elbows, while others do not admit of any movement. The bones, ligaments, cartilages, and synovial membrane, are supplied with nerves, arteries, and veins.

When an animal is provided with an internal bony structure, it indicates a high rank in the scale of organization. An elaborate texture of bone is found in no class below the vertebrates. Even in the lower order of this sub-kingdom, which is the highest of animals, bone does not exist, as is the case in some tribes of fishes, such as sharks, etc., and in all classes below that of the cartilaginous fishes, the inflexible substance which sustains the soft parts is either shell or some modification of bone, and is usually found on the outside of the body. True bone, on the contrary, is found in the interior, and, therefore, in higher animals, the skeleton is always internal, while the soft parts are placed external to the bony frame. While many animals of the lowest species, being composed of soft gelatinous matter, are buoyant in water, the highest type of animals requires not only a bony skeleton, but also a flexible, muscular system, for locomotion in the water or upon the land. Each species of the animal kingdom is thus organically adapted to its condition and sphere of life.

CHAPTER III.

PHYSIOLOGICAL ANATOMY.

THE MUSCLES.

The *Muscles* are those organs of the body by which motion is produced, and are commonly known as *flesh*. A muscle is composed of *fasciculi*, or bundles of fibers, parallel to one another. They are soft, varying in size, of a reddish color, and inclosed in a cellular, membranous sheath. Each *fasciculus* contains a number of small fibers, which, when subjected to a microscopic examination, are found to consist of *fibrillæ*, or

Fig. 19.



Muscular fibers highly magnified.

little fibers; each of these fibrillæ in turn being invested with a delicate sheath. The fibers terminate in a glistening, white *tendon*, or hard cord, which is attached to the bone. So firmly are they united, that the bone will break before the tendon can be released. When the tendon is spread out, so as to resemble a membrane, it is called *fascia*. Being of various extent and thickness, it is distributed over the body, as a covering and protection for the more delicate parts, and aids also in motion, by firmly uniting the muscular fibers. The spaces between the muscles are frequently filled with fat, which

gives roundness and beauty to the limbs. The muscles are of various forms; some are longitudinal, each extremity terminating in a tendon, which gives them a *fusiform* or spindle-shaped appearance; others are either fan-shaped, flat, or cylindrical.

Every muscle has an *origin* and an *insertion*. The term *origin* is applied to the more fixed or central attachment of a muscle, and the term *insertion* to the movable point to which the force of the muscle is directed; but the origin is not absolutely fixed, except in a small number of muscles, as those of the face, which are attached at one extremity to the bone, and at the other to the movable integument, or skin. In most instances, the muscles may act from either extremity. The muscles are

Fig. 20.



1. A spindle-shaped muscle, with tendinous terminations. 2. Fan-shaped muscle. 3. Penniform muscle. 4. Bipenniform muscle.

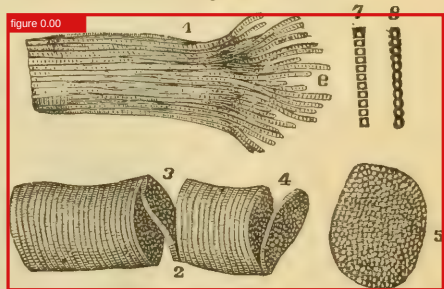
divided into the Voluntary, or muscles of animal life, and the Involuntary, or muscles of organic life. There are, however, some muscles which cannot properly be classified with either, termed Intermediate.

The *Voluntary Muscles* are chiefly controlled by the will, relaxing and contracting at its pleasure, as in the motion of the eyes, mouth, and limbs. The fibers are of a dark red color, and possess great strength. These fibers are parallel, seldom interlacing, but presenting a striped or striated appearance; and a microscopic examination

of them shows that even the most minute consist of parallel filaments marked by longitudinal and transverse *striae*, or minute channels. The fibers are nearly the same length as the muscles to which they belong. Each muscular fiber is capable of

divided into the Voluntary, or muscles of animal life, and the Involuntary, or muscles of organic life. There are, however, some muscles which cannot properly be classified with either, termed Intermediate.

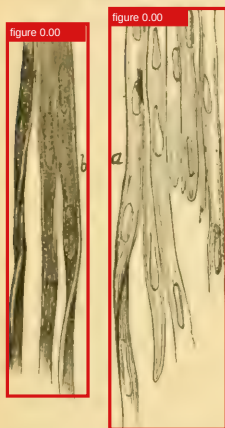
Fig. 21.



Striped muscular fibers showing cleavage in opposite directions. 1. Longitudinal cleavage. 2. Transverse cleavage. 3. Transverse section of disc. 4. Disc nearly detached. 5. Detached disc, showing the sarcous elements. 6. Fibrillae. 7, 8. Separated fibrillae highly magnified.

contraction; it may act singly, though usually it acts in unison with others. By a close inspection, it has been found that fibers may be drawn apart longitudinally, in which case they are termed *fibrillæ*, or they may be separated transversely, forming a series of discs. The *Sarcolemma*, or investing sheath of the muscles, appears to be formed even before there are any visible traces of the muscle itself. It is a transparent and delicate membrane, but very elastic. The *Involuntary Muscles* are influenced by the sympathetic nervous system, and their action pertains to the nutritive functions of the body. They differ from the voluntary muscles in not being striated, having no tendons, and in the net-work arrangements of their fibers. The *Intermediate Muscles* are composed of striated and unstriated fibers; they are, therefore, both voluntary and involuntary in their func-

Fig. 22.



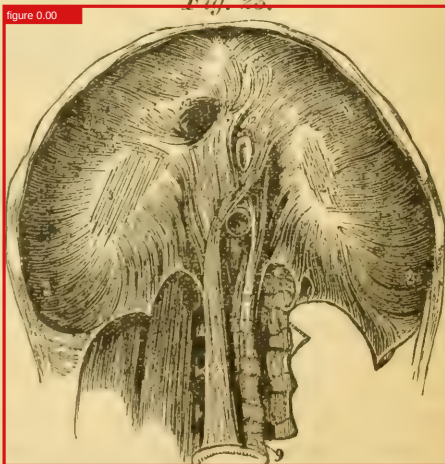
Unstriated muscular fiber; at *b*, in its natural state; at *a*, showing the nuclei after the action of acetic acid.

tions. The muscles employed in respiration are of this class, for we can breathe rapidly or slowly, and, for a short time, even suspend their action; but soon, however, the organic muscles assert their instinctive control, and respiration is resumed.

The Diaphragm, or midriff, is the muscular division between the thorax and the abdomen. It has been

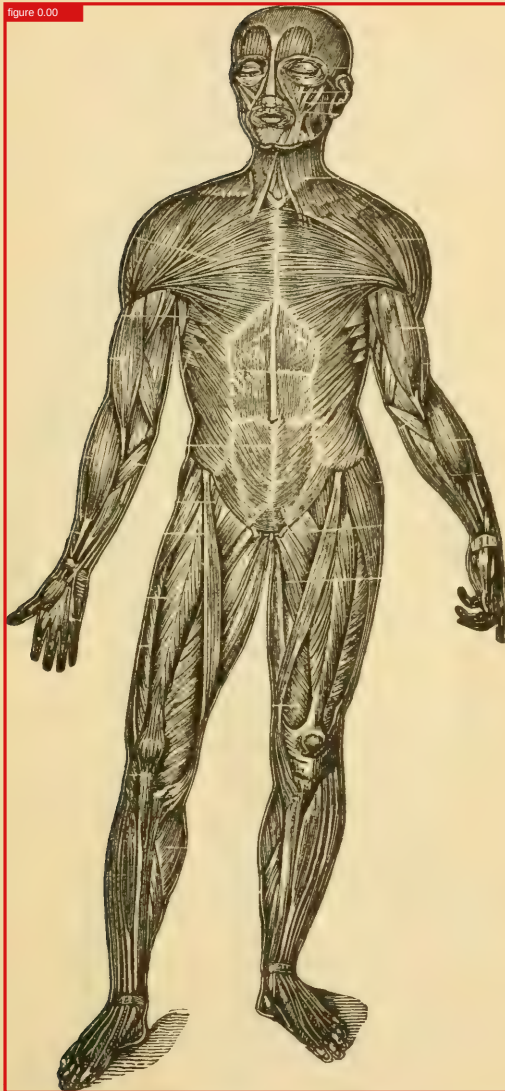
compared to an inverted basin, the concavity of which is

Fig. 23.



A view of the under side of the diaphragm.

compared to an inverted basin, the concavity of which is

Fig. 24.

A representation of the superficial layer of muscles on the anterior portion of the body.

directed toward the abdomen. The muscles receive their nourishment from the numerous blood-vessels which penetrate their tissues. The voluntary muscles are abundantly supplied with nerves, while the involuntary are not so numerously furnished. The color of the muscles is chiefly due to the blood which they contain. They vary in size according to their respective functions. For example, the functions of the heart require large and powerful muscles, and those of the eye, small and delicate ones. There are between four hundred and sixty and five hundred muscles in the human body.

Very rarely is motion produced by the action of a single muscle, but by the harmonious action of several. There is infinite variety in the arrangement of the muscles, each being adapted to its purpose, in strength, tenacity, or elasticity. While some involuntarily respond to the wants of organic life, others obey, with mechanical precision, the edicts of the will. The peculiar characteristic of the muscles is their contractility; for example, when the tip of the finger is placed in the ear, an incessant vibration, due to the contraction of the muscles of the ear, can be heard. When the muscles contract, they become shorter; but what is lost in length is gained in breadth and thickness, so that their actual volume remains the same. Muscles alternately contract and relax, and thus act upon the bones. The economy of muscular power thus displayed is truly remarkable. In easy and graceful walking, the forward motion of the limbs is not altogether due to the exercise of muscular power, but partly to the force of gravity, and only a slight assistance of the muscles is required to elevate the leg sufficiently to allow it to oscillate.

Motion is a characteristic of living bodies. This is true, not only in animals, but also in plants. The oyster, although not possessing the power of locomotion, opens and closes its shell at pleasure. The coral insect appears at the door of its cell, and retreats at will. All the varied motions of animals are due to a peculiar property of the muscles, termed *contractility*. Although plants are influenced by external agents, as light, heat, electricity, etc., yet it is supposed that they may move in response to inward impulses. The sensitive stamens of the barberry, when touched at their base on the inner side, resent the intrusion, by making a sudden jerk forward. Venus's

Fig. 25.

A representation of the superficial layer of muscles on the posterior portion of the body.

fly-trap, a plant found in North Carolina, is remarkable for the sensitiveness of its leaves, which close suddenly and capture insects which chance to alight upon them. The muscles of the articulates are situated within the solid frame-work, unlike the vertebrates, whose muscles are external to the bony skeleton. All animals have the power of motion, from the lowest radiate to the highest vertebrate, from the most repulsive polyp to that type of organized life made in the very image of God.

The muscles, then, subserve an endless variety of purposes. By their aid the farmer employs his implements of husbandry, the mechanic deftly wields his tools, the artist plies his brush, while the fervid orator gives utterance to thoughts glowing with heavenly emotions. It is by their agency that the sublimest spiritual conceptions can be brought to the sphere of the senses, and the noblest, loftiest aims of to-day can be made glorious realizations of the future.

CHAPTER IV.

PHYSIOLOGICAL ANATOMY.

THE DIGESTIVE ORGANS.

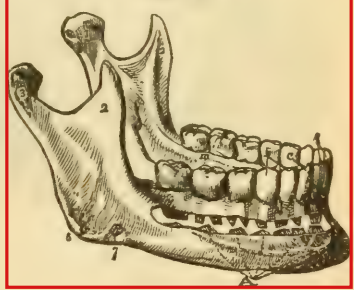
Digestion signifies the act of separating or distributing, hence its application to the process by which food is made available for nutritive purposes. The organs of digestion are the Mouth, Teeth, Tongue, Salivary Glands, Pharynx, Esophagus, the Stomach and the Intestines, with their glands, the Liver, Pancreas, Lacteals, and the Thoracic Duct.

The *Mouth* is an irregular cavity, situated between the upper and the lower jaw, and contains the organs of mastication. It is bounded by the lips in front, by the cheeks at the sides, by the roof of the mouth and teeth of the upper jaw above, and behind and beneath by the teeth of the lower jaw, soft parts, and palate. The soft palate is a sort of pendulum attached only at one of its extremities, while the other involuntarily opens and closes the passage from the mouth to the pharynx. The interior of the mouth, as well as other portions of the alimentary canal, is lined with a delicate tissue, called *mucous membrane*.

The *Teeth* are firmly inserted in the alveoli or sockets, of the upper and the lower jaw. The first set, twenty in number, are temporary, and appear during infancy. They are replaced

figure 0.00

Fig. 26.



A view of the lower jaw. 1. The body. 2. 2. Rami, or branches. 3, 3. Processes of the lower jaw. m. Molar teeth. b. Bicuspids. c. Cuspids. i. Incisors.

by permanent teeth, of which there are sixteen in each jaw ; four incisors, or front teeth, four cuspids, or eye teeth, four bicuspids, or grinders, and four molars, or large grinders. Each tooth is divided into the crown, body, and root. The *crown* is the grinding surface ; the *body*, the part projecting from the jaw, is the seat of sensation and nutrition ; the *root* is that portion of the tooth which is inserted in the alveolus. The teeth are composed of dentine, or ivory, and enamel. The ivory forms the greater portion of the body and root, while the enamel covers the exposed surface. The small white cords communicating with the teeth are the nerves.

The *Tongue* is a flat oval organ, the base of which is attached

Fig. 27.



The salivary glands. The largest one, near the ear, is the parotid gland. The next below it is the submaxillary gland. The one under the tongue is the sublingual gland.

to the os hyoides, while the apex, the most sensitive part of the body, is free. Its surface is covered with a membrane, which, at the sides and lower part, is continuous with the lining of the mouth. On the lower surface of the tongue, this membrane is thin and smooth, but on the upper side it is covered with

numerous papillæ, which, in structure, are similar to the sensitive papillæ of the skin.

The *Salivary Glands* are six in number, three on each side of the mouth. Their function is to secrete a fluid called *saliva*, which aids in mastication. The largest of these glands, the *Parotid*, is situated in front and below the ear ; its structure, like that of all the salivary glands, is cellular. The *Submaxillary* gland is circular in form, and situated midway between the

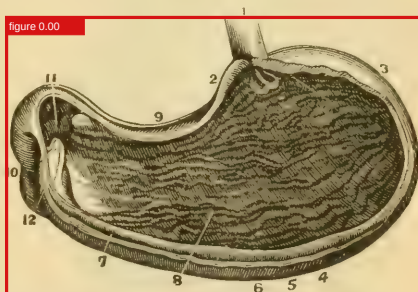
angle of the lower jaw and the middle of the chin. The *Sublingual* is a long flattened gland, and, as its name indicates, is located below the tongue, which when elevated, discloses the saliva issuing from its porous openings.

The *Pharynx* is nearly four inches in length, formed of muscular and membranous cells, and situated between the base of the cranium and the esophagus, in front of the spinal column. It is narrow at the upper part, distended in the middle, contracting again at its junction with the esophagus. The pharynx communicates with the nose, mouth, larynx, and esophagus.

The *Esophagus*, a cylindrical organ, is a continuation of the pharynx, and extends through the diaphragm to the stomach. It has three coats: first, the muscular, consisting of an exterior layer of fibers running longitudinally, and an interior layer of transverse fibers; second, the cellular, which is interposed between the muscular and the mucous coat; third, the mucous membrane, or internal coat, which is continuous with the mucous lining of the pharynx.

The *Stomach* is a musculo-membranous, conoidal sac, communicating with the esophagus by means of the cardiac orifice (see Fig. 28). It is situated obliquely with reference to the body, its base lying at the left side, while the apex is directed toward the right side. The stomach is between the liver and spleen, subjacent to the diaphragm, and communicates with the intestinal canal by the pyloric orifice. It has three coats. The peritoneal, or external coat is composed of compact, cellular tissue, woven into a thin, serous membrane, and assists in keeping the stomach in place. The middle coat is formed of three layers of muscular fibers: in the first, the fibres run

Fig. 28.



A representation of the interior of the stomach.
 1. The esophagus. 2. Cardiac orifice opening into the stomach. 3. The middle or muscular coat. 4. The interior or mucous coat. 5. The beginning of the duodenum. 6. The pyloric orifice. 7. The interior or mucous coat. 8. The beginning of the duodenum. 9. The interior or mucous coat. 10. The beginning of the duodenum. 11. The pyloric orifice. 12. The interior or mucous coat.